Observing CO in the channel

Absorption, atmospheric separation and the role of path length

60–1000 GHz

14 October 2026



The channel is a frequency × length matrix

$$H_{ij} = h(f_i, l_j)$$

f samples frequency
l samples atmospheric path length

Each new column needs a different path length.

	l_1	l_2	l_3
f_1	h_{11}	h_{12}	h_{13}
f_2	h_{21}	h_{22}	h_{23}
f_3	h_{31}	h_{32}	h_{33}

Current case: one gas, no receiver noise
Later: other gases, mixtures, then angle

Geometry and absorption contribute to one channel

$$|h| = |h_{fs}| \exp(-\tau_m / 2) \exp(-\tau_c / 2)$$

Term	Meaning
hFS	Geometric amplitude gain, dependent on frequency and distance
τm	Optical depth of the background atmosphere
τc	Additional optical depth due to the chemical cloud

Amplitude transmission: $\exp(-\tau / 2)$ Power transmission: $\exp(-\tau)$

A matched reference isolates the chemical loss

h_0: the same channel without the added CO

$$|a_c| = |h| \,/\, |h_0|$$

$$\tau_c = -2 \ln(|h| \,/\, |h_0|)$$

The reference preserves geometry, atmosphere and instrument response.

An unknown atmosphere leaves the split ambiguous

$$r = -2 \ln(|h| / |h_{fs}|) = \tau_m + \tau_c$$

Two decompositions

$$\tau_m + \tau_c$$

$$(\tau_m + q) + (\tau_c - q)$$

Both produce the same channel.

Separation needs a matched reference or distinguishable model responses.

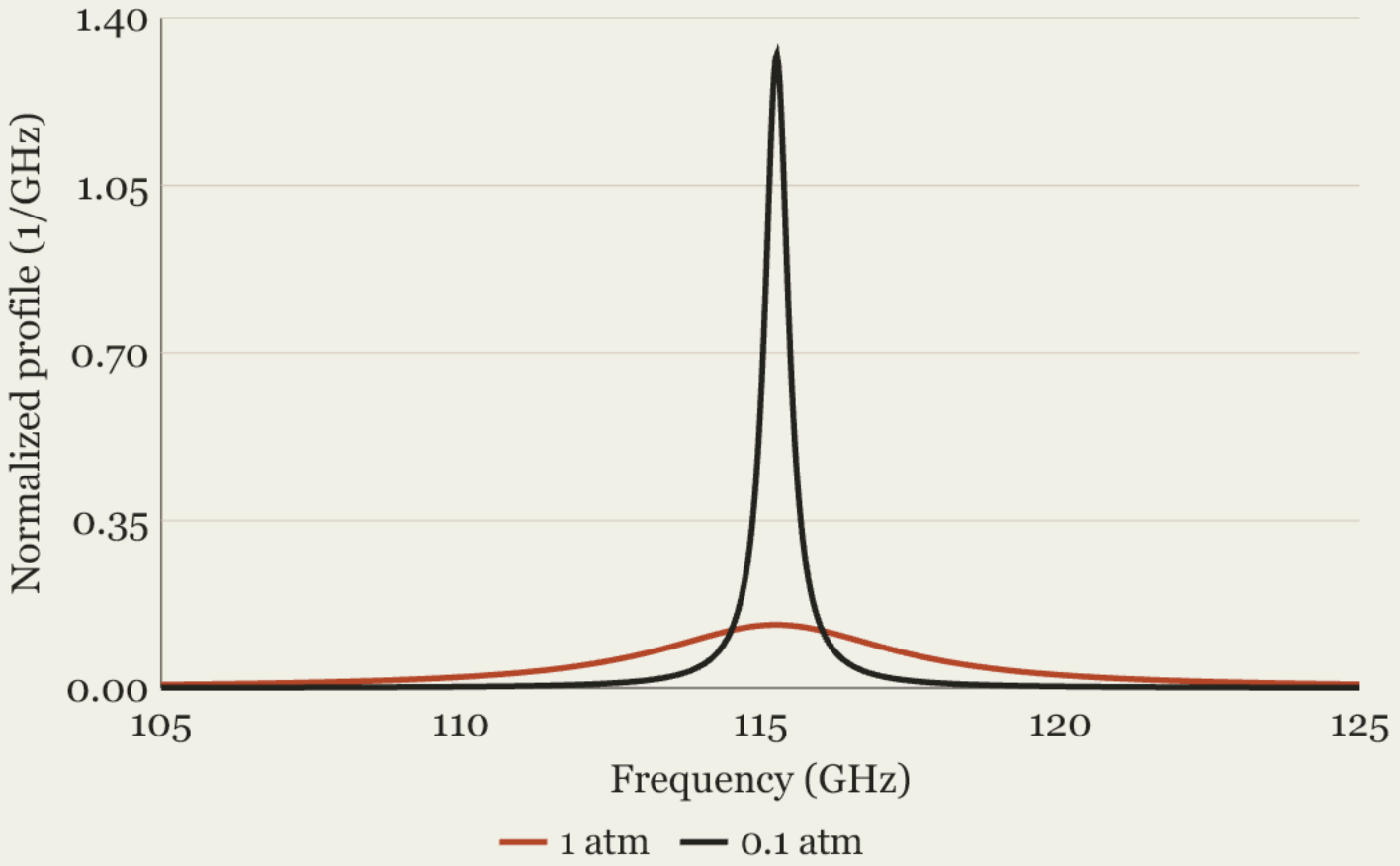
A discrete transition has a continuous profile

$$f_0 = \Delta E / h_P$$

Collisions shorten molecular coherence.

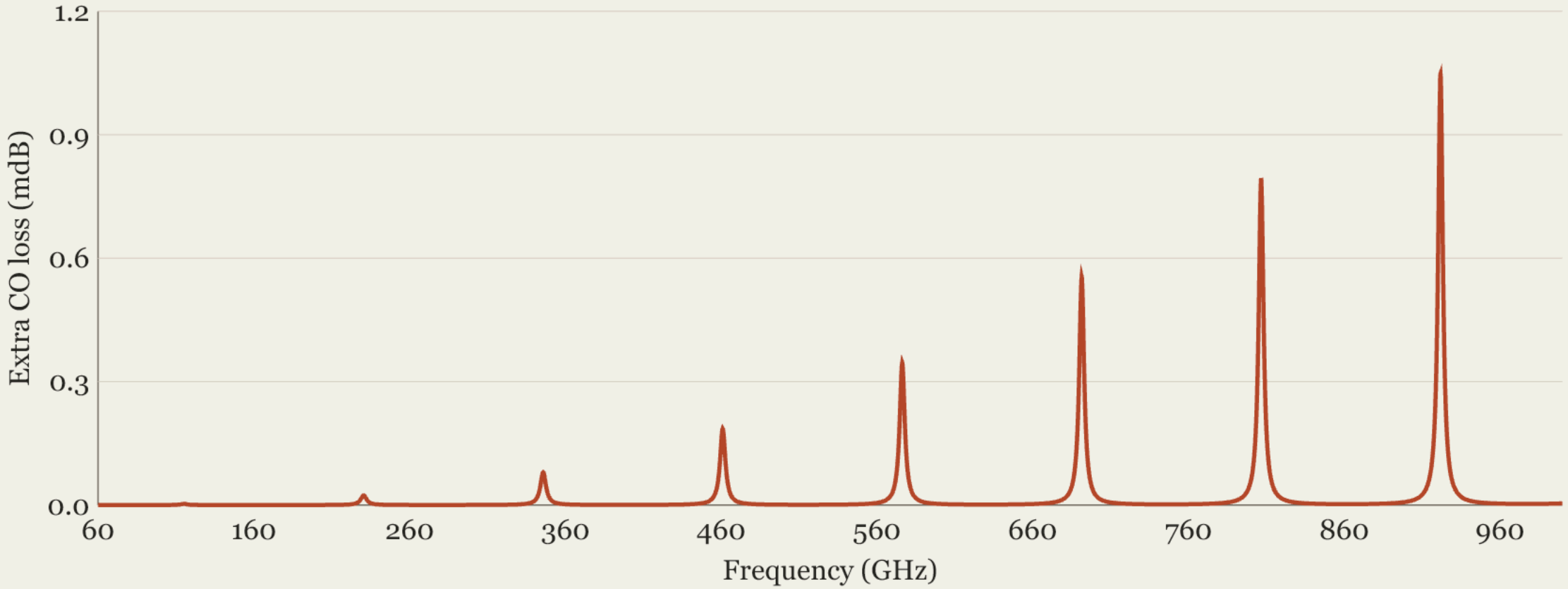
Thermal motion introduces Doppler shifts.

At 1 atm, this line has a Lorentz full width of about 4.82 GHz.



CO at 296 K. Equal integrated profile area. Source: HITRAN.

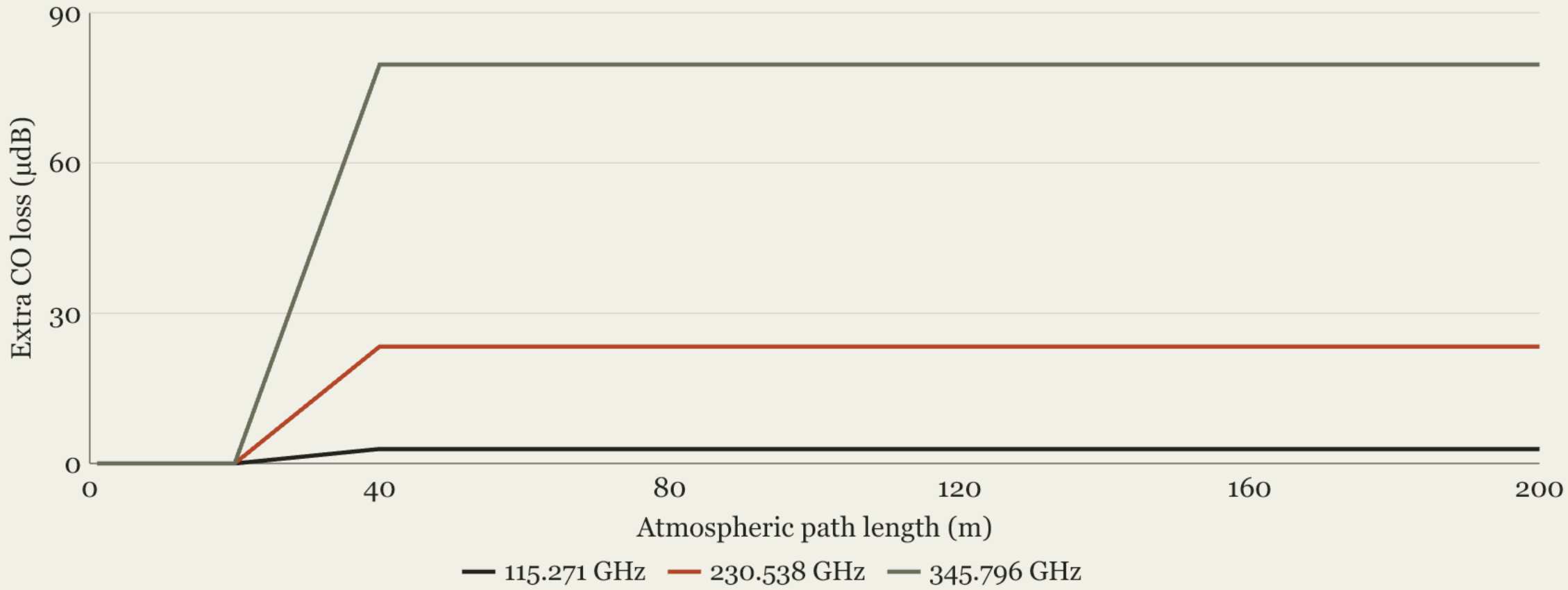
CO has eight main rotational lines in this band



115.271 230.538 345.796 461.041 576.268 691.473 806.652 921.800 GHz

Predicted extra loss for 1 ppm over 20 m at 296 K and 1 atm. Main isotopologue.

Chemical loss grows only while crossing the cloud



Before 20 m: no added CO 20–40 m: column grows After 40 m: same CO column

Nested paths through one stationary medium. Excess CO lies between 20 and 40 m.

A chosen CO column gives a very small channel change

1 ppm of excess CO across a 20 m cloud

296 K and 1 atm
Water partial pressure: 1000 Pa
Total path in this comparison: 200 m

Frequency	Extra loss	|h| / |ho|
115.271 GHz	2.89 µdB	0.9999996673
230.538 GHz	23.35 µdB	0.9999973120
345.796 GHz	79.66 µdB	0.9999908288

The matched reference recovers the input column under the same model.

Conditional simulation. Exact atmosphere and calibration. No receiver noise.

The separation requires more than an absorption law

Information	Role in the separation
Geometry and instrument response	Separate path loss and instrument gain
Temperature, pressure and humidity	Set background loss and molecular line shapes
Reference CO and gas spectra	Separate target enhancement from other gases
Cloud length or spatial model	Convert molecular column to concentration

A standard atmosphere supplies assumed conditions. A matched reference must reproduce the conditions of the sample.

Chemical identifiability and practical detectability

Identifiability

Can the gas produce a spectral response that the allowed background and other gases cannot reproduce?

Practical detectability

Is the resulting channel change measurable at the chosen column, path length and instrument precision?

The noiseless study addresses the first question. A useful concentration limit requires an error model.

The next experiment needs a defined reference

How will we obtain ho?

A matched measurement
or a constrained atmospheric model

Then vary the uncertain atmospheric quantities
and test whether CO remains distinguishable.

Appendix: the molecular model

Transition positions

$$E_J = h_P \, B \, J(J + 1)$$

CO rigid rotor. J increases by one in absorption.

Cross section

$$\sigma_c(f) = \Sigma_j \, S_j \, V_j(f - f_j)$$

Voigt combines Lorentz and Doppler broadening.

Propagation

$$\tau_c = \int n_c(s) \, \sigma_c(f; s) \, ds$$

$$|a_c| = \exp(-\tau_c / 2)$$

For uniform conditions: $\tau c(f, l) = \sigma c(f) \, Nc(l)$, where Nc is the molecular column.

Sources and reproducible calculation

HITRAN	Line positions, strengths and broadening
JPL CO catalog	Independent rotational line positions
ITU P.676-13, Annex 1	Atmospheric gas attenuation to 1000 GHz
NIST spectroscopy	Line broadening and molecular energy levels
Project calculation	CO spectra, channel matrices and assumptions

Complete derivations and source links accompany each slide in the speaker notes.